

Lab 1: Introduction to Java Fundamentals (Variables, Input, Methods & Arrays)

1. Objective

Objective: The purpose of this lab is to understand basic Java syntax, handle runtime user input, modularize code using methods, and manage data effectively using loops and arrays.

Task 1: Variables and User Input (Scanner)

Variables are used to store data in memory, and the Scanner class is used to take input from the user at runtime.

```
import java.util.Scanner;

public class Lab1_Task1 {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter your name: ");

        String name = scanner.nextLine();

        System.out.print("Enter your age: ");

        int age = scanner.nextInt();

        System.out.println("Welcome, " + name + "! You are " + age + " years old.");

        scanner.close();

    }

}
```

Task 2: Using Methods (Functions)

In OOP, methods are used to divide code into smaller, reusable blocks to keep code clean and modular.

```
public class Lab1_Task2 {
    // Method to calculate the sum of two numbers
    public static int addNumbers(int a, int b) {
        return a + b;
    }

    public static void main(String[] args) {
        int num1 = 15;
        int num2 = 25;
        int result = addNumbers(num1, num2);

        System.out.println("Sum is: " + result);
    }
}
```

Task 3: Loops And Conditionals

Use for loops to repeat tasks and if-else statements to check conditions.

```
public class SimpleLoopExample {
    public static void main(String[] args) {
        // Printing numbers from 1 to 5 using a for loop
        for (int i = 1; i <= 5; i++) {
            System.out.println("Current number: " + i);
        }
    }
}
```

Task 4: Reverse Loop

```
public class ReverseLoopExample {
    public static void main(String[] args) {
        // Printing numbers from 5 down to 1 using a for loop
        for (int i = 5; i >= 1; i--) {
            System.out.println("Current number: " + i);
        }
    }
}
```

Task5: Skip Numbers-

```
public class SkipNumberLoopExample {
    public static void main(String[] args) {
        System.out.println("Printing numbers from 1 to 20, skipping one number each
time:");
        // i += 2 increases the counter by 2 instead of 1
        for (int i = 1; i <= 20; i += 2) {
            System.out.println("Current number: " + i);
        }
    }
}
```

Task6: MultiplicationTableTask:

```
import java.util.Scanner;
public class MultiplicationTableTask {
    public static void main(String[] args) {
        // Create a Scanner object to take input from the user
        Scanner scanner = new Scanner(System.in);
        // Ask the user to enter an integer
        System.out.print("Enter an integer to print its multiplication table: ");

        int number = scanner.nextInt();

        System.out.println("\n--- Multiplication Table for " + number + " ---");
        // Use a for loop to print the table from 1 to 10
        for (int i = 1; i <= 10; i++) {
            int result = number * i;

            System.out.println(number + " x " + i + " = " + result);
        }
    }
}
```

Task7: Write a Factorial Program :

```
import java.util.Scanner;

public class FactorialCalculator {

    public static void main(String[] args) {

        // Create a Scanner object to take input from the user

        Scanner scanner = new Scanner(System.in);

        // Ask the user to enter a number

        System.out.print("Enter a positive integer to find its factorial: ");

        int number = scanner.nextInt();
        // Use long because factorial values become very large quickly
        long factorial = 1;
```

```

// Check if the number is negative
if (number < 0) {

    System.out.println("Factorial is not defined for negative numbers.");

} else {

    // Loop from 1 up to the entered number
    for (int i = 1; i <= number; i++) {

        factorial *= i; // This is short for: factorial = factorial * i;

    }
    // Print the final result
    System.out.println("The factorial of " + number + " is: " + factorial);

}    } }

```

Task8: Basic Program: while Loop:

A while loop checks the condition before running the code inside it. If the condition is true, the loop runs; if it's false, the loop doesn't run at all.

```

public class WhileLoopExample {

    public static void main(String[] args) {

        // Step 1: Initialize the counter variable outside the loop

        int i = 1;

        System.out.println("Counting from 1 to 5 using a while loop:");

        // Step 2: The condition is checked before running the block

        while (i <= 5) {

            System.out.println("Current number: " + i);

            // Step 3: Increment the counter so the loop eventually stops

            i++;

        }

    }

}

```

Task9: Basic Program: do-while Loop:

A do-while loop is different because it runs the code block at least once before checking the condition at the end.

```
import java.util.Scanner;

public class DoWhileLoopExample {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        int number;

        // The 'do' block runs at least once, regardless of the condition
        do {

            System.out.print("Enter a number (enter 0 to exit): ");

            number = scanner.nextInt();

            System.out.println("You entered: " + number);

        } while (number != 0); // Condition is checked AFTER the block executes

        System.out.println("Program stopped because you entered 0.");

        scanner.close();

    }

}
```

Task10: What is a for-each Loop?

The for-each loop (also known as the Enhanced For Loop) in Java is specially designed to loop through items in arrays or collections (like lists).

Unlike a regular for loop, you don't need to worry about tracking index numbers (i), checking lengths, or incrementing counters manually. It automatically goes through each element from the beginning to the end, making your code much cleaner and easier to read.

Basic Syntax :

```
for (DataType item : arrayName) {  
    // code using 'item'  
}
```

Task11: String Array (Listing Items)

This program uses a `for-each` loop to print a list of fruit names from a `String` array.

```
public class ForEachExample2 {  
  
    public static void main(String[] args) {  
  
        // An array of string values  
  
        String[] fruits = {"Apple", "Banana", "Mango", "Orange", "Grapes"};  
  
  
        System.out.println("List of available fruits:");  
  
  
        // For-each loop: 'fruit' takes the value of each string in the array  
  
        for (String fruit : fruits) {  
  
            System.out.println("I like eating " + fruit);  
  
        }  
  
    }  
}
```

Task12: For Each Loop

```
public class ArrayInitializationExample {

    public static void main(String[] args) {

        // =====

        // 1. Static Initialization (Values known beforehand)

        // =====

        // We declare the array and assign values directly inside curly braces { }

        int[] numbers = {10, 20, 30, 40, 50};

        System.out.println("--- Static Initialization ---");

        System.out.println("Printing elements using a for-each loop:");

        for (int num : numbers) {

            System.out.print(num + " ");

        }

        // =====

        // 2. Dynamic Initialization (Size specified, values assigned later)

        // =====

        // We create an array of size 3 using the 'new' keyword

        int[] scores = new int[3];

        // Assigning values to specific index positions manually

        scores[0] = 85;

        scores[1] = 90;

        scores[2] = 95;

        System.out.println("\n\n--- Dynamic Initialization ---");
```

```
System.out.println("Printing elements using a for-each loop:");
```

```
    for (int score : scores) {  
  
        System.out.print(score + " ");  
  
    }  
  
}  
  
}
```

PRACTICE TASKS FOR STUDENTS !

Complete the following 5 tasks to test your understanding:

Task1: Write a method Without parameters & without return type (void)

Task2: With parameters & without return type (void)

Task3: With parameters & with return type (showing both storing in a variable and direct printing)

Task4: Write A program which prints the Elements of an array !

Task5: Write a program which prints Sum of an array Using the For each Loop !

Task6: Create a method in java that can accept any number of integer arguments and returns the sum of all those numbers. Use varargs and for-each loop.

Task7: Write a program That loop will run infinite : write program in two ways :in both ways use for loop !

